Seminar, Practical Work, Thesis at the Institute for Machine Learning

Last changes: 27 Aug 2026

Please self-enroll in this Moodle course.

For a general explanation, watch this recording of the “Intro and Q&A” session (WS25).



Contents:

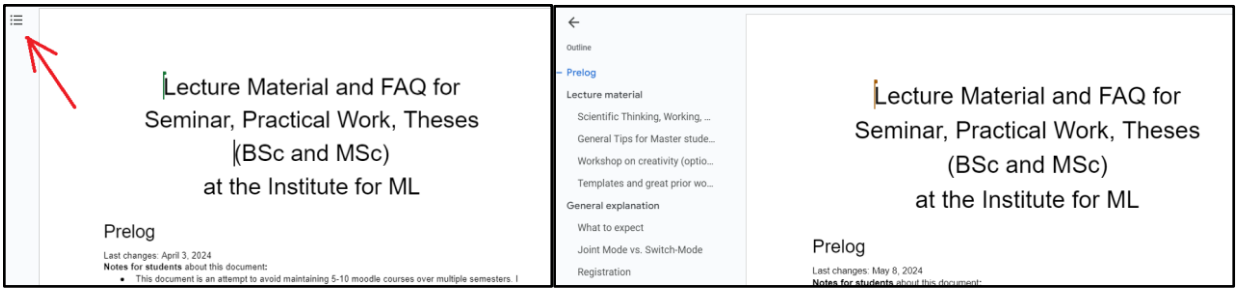
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Prologue

About this document

- This document collects all essential information in one place to avoid maintaining multiple Moodle pages over multiple semesters.
- There is just one **official Moodle page** for these SPT courses at IML. Any older versions (before SS24) are obsolete; **this document is the authoritative source**.
- You're encouraged to **make comments** for clarification—it's better to answer once here than reply to 20 emails. **Please don't resolve comments**, so others can benefit.
- If you have improvement suggestions, please let us know.
- Feel free to share this doc with peers (e.g., students who do their SE/PW/TH at other institutes).
- Numbered lists are ordered (usually step-by-step guides or priorities); bullet points are unordered.
- The document outline on the left is handy for navigating through the document.



Contact info

- For de/registration: IML Office office@ai-lab.jku.at

- If there is a question that is not covered here and not meant for a broad audience, please contact one of us (pick the one based on the first letter of your last name):
    - A, B, C, D, E, F, G, H, G, I, J, K, L, Andreas Radler radler@ml.jku.at
    - M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z Sohvi Luukkonen luukkonen@ml.jku.at

Important: for **any request via mail**, please always be specific and add:

- Your matriculation number (e.g., K12345678)
- The LVA-Nr from KUSSS (e.g., 365.279) and the semester (e.g., WS23 or SS24)
- Your supervisor's name (if you have a supervisor already)

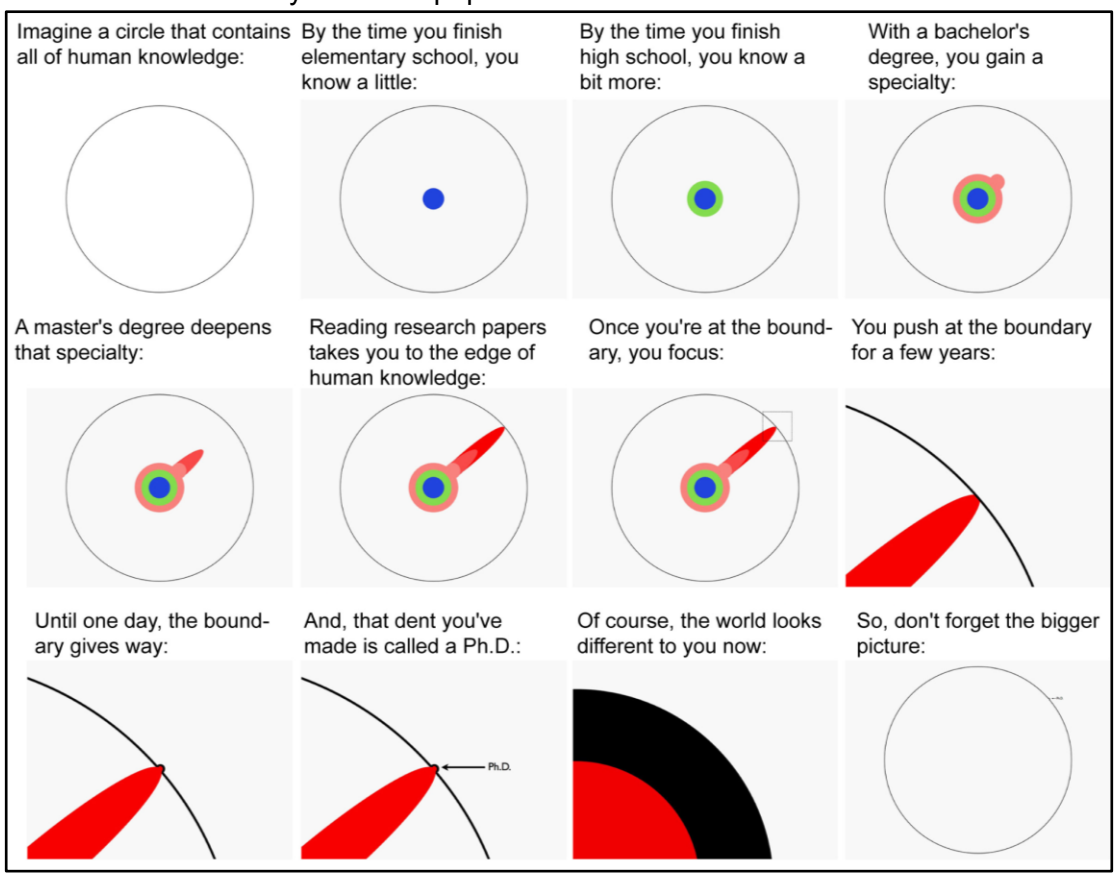
General explanation

Why do we offer these courses?

These courses shall help you improve your skills around ‘how to think and work scientifically’. Have a look at the mental model for “the goal of a PhD” below.
Intuitively, you shall learn to “push yourself to the boundary of human knowledge” and “apply the scientific method” in a “sandbox setup”. Concretely, you should get into the habit of:

- **Understand important context about science:** Understand how scientific progress is made, how new knowledge is created, distributed, and evolving.
- **Orient yourself better in the literature landscape:** Think in scientific fields, where to find the right papers, how to critically reflect on papers, etc
- **Read and reproduce state-of-the-art papers**
- **Critically reflect on the papers**
- **Understand the scientific method** and apply it to simple problems (read literature, form hypotheses, experiment, evaluate, …). For more info, also see the scientific method section in the FAQ below.

The goal of a PhD (source)
Note that the goal of a BSc/MSc is to prepare you to potentially do a PhD afterwards. The expectation is not to actually do a real paper.



What to expect?

There are 7 courses in total that shall help you progress in Scientific Working. You can find more details below in the courses section.

- **Seminar in AI (BSc or MSc)**: The goal is to get you started with paper-reading. So you will have to read a paper (and probably some related papers more superficially). Then you will have to present this paper and lead a Q&A session, where your supervisor and the audience will ask you questions. The presentation shall be according to scientific standards. Most importantly, you use references. Notice that this is roughly step 1 of the scientific method (see below)
    - BSc vs. MSc: The requirements for an MSc are more independent, with broader literature work and a more critical analysis.
- **Practical Work in AI (BSc or MSc)**: The goal is to get you started with (re-)implementing methods from papers. This not only deepens your understanding of a topic, but it also practices your coding skills and is a first step towards the “scientific method”.
    - Notice that these are roughly steps 2-4 of the scientific method
- **Bachelor's Thesis Seminar**: The goal is to do a first attempt at a scientific project. I.e., applying the scientific method in a smaller, more controllable setting than a bigger research project.
- **Master's Thesis Seminar**: A presentation of your master's thesis (the poster session).

BSc vs. MSc

Seminar and Practical work are very similar for BSc and MSc. In the Seminar, you have to present a paper, and in the Practical work, you have to implement something (usually reproduce one or two methods on one or two datasets and compare them).

For the thesis, the courses in the BSc and MSc differ slightly. For the BSc, you just need to be registered for the Bachelor’s Thesis Seminar course to get your grade for the BSc Thesis. The Master's Thesis Seminar course will be the grade for your presentation in a poster session. It’s NOT your grade for the master's thesis itself (see the FAQ section for more details).

Same-topic Mode vs. Switch-topic Mode

- “**Same-topic mode**” means to do the Seminar, Practical Work, and the Thesis on the same topic with the same supervisor at the same institute
- “**Switch-mode**” means to switch topics/supervisors/institutes between Seminar, Practical Work, or Thesis.

At the IML, we usually think the Same-topic mode makes more sense, since there are more synergies. Still, there’s flexibility if you want to switch. The important part is always to have a supervisor who wants to supervise you. Topics and Registration

Important: If you were registered for a course in a previous semester, do NOT register for it once more in the new semester. E.g., student A registered for the “Seminar in AI (BSc)” in SS24, but was not able to complete the course. It is not necessary to register again for “Seminar in AI (BSc)” in SS25.

Topics and registration

Phase 0: Register in KUSSS

1. First, find a supervisor, then register for the course at the supervisor’s institute
    a. If you have not found a topic and supervisor at the start of the semester, don’t panic! Keep searching for a supervisor at the IML or other institutes (see topic list below).
    b. Being registered in KUSSS does not guarantee you a spot. Also, it does not matter if you are on the waiting list or “in the course”. The important point is that you have a topic and a supervisor. To reduce our administrative burden, we will add everyone on the waiting list to the course at the start of the semester. Again, this does not guarantee that you can do the course at our institute. So what counts is that you find a topic and a supervisor.
    c. It’s fine if you start late or find a supervisor in the subsequent semester. We would then register you for the latest course. However, **you must find a topic and a supervisor first.** Note that the deadlines are not extended. Therefore, if you start very late or have delays, you might have less time to write the report.
2. If you have found a topic and supervisor already, please register for the KUSSS course of the supervisor’s institute. For the IML, please contact the secretary (see contact section above). If you have done this already, you can stop here.

Phase 1: Find a topic and a supervisor

1. Please look through the topic list(s) below. Also, check the topic lists from other institutes.
    a. Institute for Machine Learning: TOPIC LIST (usually this list is updated at the beginning of each semester)
    b. OVERVIEW of ALL INSTITUTES for AI SE/PW/TH: TOPIC LISTS OF ALL ML INSTITUTES.
2. Write a supervisor and ask whether they would supervise you. The Seminars, Practicals and Thesis for AI students mail addresses are on the other tab of the topic list.
    a. Please don’t expect an answer within 3 days. If there is no answer within a week, please forward one of us (see contact info) the email and explain that the inquiry to the supervisor remains unanswered. We will then ping that person.
3. ONLY if you have found a supervisor and they agree to supervise you, go to Phase 2

Phase 2: Deregister in KUSSS at the old institute, then register for the right course in the new institute

Assumption: This assumes the semester has already started. A cut-off date after the semester has started, you can not de/register yourself from a KUSSS course. Therefore, you have to ask at the institute directly if you want to register/deregister (in our case, office@ai-lab.jku.at).

1. If you are currently registered for a KUSSS course at another institute, deregister there first.
2. If you have a topic and supervisor at the IML, you should register for our KUSSS course(s). For this, write an email to office@ai-lab.jku.at with your full name, immatriculation number, and the course number(s)* you want to be registered for. Also, mention which topic you will do and who your supervisor is. If in KUSSS the IML offers several courses with the same name, it does not matter which course you are registered for.
3. Our colleagues will then put you into the KUSSS course(s).

*See the correct course numbers in the Course-specific information of each course. Your supervisor doesn't need to be on the list of lecturers.

Course material (mandatory)

Lectures in WS26

The lectures will be in hybrid mode, and recordings will be made available. So you may come to the institute, or you may join virtually. You either have to attend or watch the recording (it will help you execute these courses) at the beginning of your SE/PW/TH journey at IML.

You must attend or watch each lecture at least once and, starting from WS26, complete the corresponding Moodle quizzes to be able to complete any of the SPT courses at IML.

- Intro and Q&A
  - Date: Mo. 05.10.2026 12:00 - 13:30
  - Zoom link
  - Room: MT 128
  - Responsible: Andreas Radler
  - Recording SS26
- Intro to Scientific Thinking and Working + Project- and Self-Management Advice
  - Dates & Rooms:
    - Fri 16.10.2026 10:15 - 11:45 (MT 127)
    - Fri 23.10.2026 10:15 - 11:45 (MT 127)
  - Zoom link (for both dates)
  - Responsible: Andreas Radler
  - Slides (scientific thinking)
  - Slides (project- and self-management)
  - Recording Session 1 WS 26
  - Recording Session 2 WS 26
  - ~~Recording SS26~~ (outdated! There are new slides this semester!)
- Introduction to Scientific Writing and Reading
  - Dates & Rooms
    - Fri 30.10.2026 10:15 - 11:45 (MT 127)
    - Fri 06.11.2026 10:15 - 11:45 (S2 044 in SCP2)
  - Zoom link (for both dates)
  - Responsible: Sohvi Luukkonen
  - Recording SS26 (Part 1)
  - Recording SS26 (Part 2)
  - Slides (SS26)

- Poster session for Master's Thesis Seminar: see the corresponding course in the **specific course section**
- For Seminar presentations, see the corresponding course in the **specific course section**

Templates, examples, and other resources

- Templates and great reports/theses of prior students (includes JKU presentation template and others)
- Official JKU LaTeX templates for reports and theses
    - For a master's thesis, you must use the official JKU template.
    - For the rest, you are free to choose any suitable template (for example, the NeurIPS template)
- Generic poster templates: Gemini template (not the chatbot), Overleaf templates, [please add here if you find a nice one]
- For seminar presentations, please stick roughly to the structure mentioned in this template (especially, please use a clear outline to avoid getting lost in slides)
- Links to prior presentations
    - Seminar in AI (MSc) presentation video from Abdulkader
    - Seminar in AI (MSc) presentation video from Dimitrios

Course-specific information

Seminar in AI (BSc)

KUSSS IDs: 365.252, 365.278, 365.370, 365.371, 320.003
Summer Semester

Prequisit

- This is a 4th-semester course; we expect to have done most of the following courses
    - Hands-on AI I and II
    - Programming in Python I and II
    - Math for AI I, II, and III
    - ML: Basic & Supervised Techniques

Goals

- Get started/practice reading papers
- Establish connections between publications
- Learn some basics for scientific presentations/reports (e.g., cite properly)

Lectures

- It is mandatory to watch these lectures at least once

Work load

- 3 ECT: ~75h (~ 2 weeks full time)

Output

- **Presentation**
    - **Format:** **In-person** (Linz or Vienna). Virtual only by approved email exception (e.g., students based in Bregenz).
    - **Task**: Present a paper plus your critical assessment.
    - **Resources:**
        - templates section, for example, presentations, PowerPoint templates
        - grading guidelines for expected content
    - **Time:** 15 mins + 5 mins Q&A. Practice to stay ≤15 min.
    - **Registration:** Registration will open approximately 2 weeks before the presentation dates. Please register for the Moodle course at the top to receive announcements. **Send your report draft** (with story-summary questions answered) to your supervisor for permission to present at least 2 weeks before. Expect a reply within 5 days (send a polite reminder if needed). **After approval**, register for the presentation at least 1 week in advance.
    - **Preparation and feedback:** Send the slide draft to your supervisor at least 1 week in advance; expect feedback at least 2 days prior (send a polite reminder if needed).
    - **Presence and peer review:** You must attend all presentations in your session (i.e., talks hosted by the same chair) and provide feedback to the presenters.
    - **Submission:** Upload slides by **23:59** the day before (see file uploads).
- **Report**

    - **Contents:** Short story summary (see this section) + paper summary and critical analysis.
    - **Length/structure:** Follow grading guidelines for expected content and word limits.
    - **Deadline:** End of the subsequent semester (e.g., start SS25 → due end WS25 ≈ 28 Feb 2026; see official semester dates here).
    - **Recommendation:** Wrap up within a few weeks after your presentation.

Grading

- Seminar Grading Guidelines.
- Once you have given your presentation and submitted the report, fill out the grading request. The grade should be available in KUSSS within a month.

Dates SS26

IMPORTANT: Please reserve the whole time range until your specific time slot is fixed. Registration will be announced and shared via Moodle and will open approximately 2 weeks before the presentation dates. Please register for the Moodle course at the top to receive announcements.

- **Seminar presentation dates (first dates).**
    - 22. May 2026: 09:00-19:00
    - Rooms: Linz S3 0318 / S2 059 & Vienna S1 / S4
    - Schedule (preliminary)
- **Seminar presentation dates (second dates)**
    - 17. July 2026: 09:00-19:00
    - Rooms: Linz S3 057 & Vienna S1
    - Schedule TBA

Practical Work in AI (BSc)

KUSSS IDs: 365.266, 365.267
Winter Semester

Prequisit

- This is a 5th-semester course; we expect to have done most of the following courses
    - Hands-on AI I & II
    - Programming in Python I & II
    - Math for AI I, II & III
    - ML: Basic, Supervised & Unsupervised Techniques

Lectures

- It is mandatory to watch these lectures at least once

Work load

- 7.5 ECT: ~188h (~5 weeks full time)

Grading

- Practical Work Grading Guidelines
- No presentation!
- Report

    - The word limit is in the grading guidelines
    - Report deadline: End of the subsequent semester.
        - E.g., if you start in SS25, the report is due at the end of WS25 (i.e., around 1st of February 2026). Semester end dates are taken from here
        - Once the report and the grading request have been submitted, the grade should be available in KUSSS within a month.
- Code: either add a link to a GitHub repo or upload a separate zip file. (Please don't put the code together with the report into a zip. We would really like to have the report as a separate PDF.)

Bachelor's Thesis Seminar in AI

KUSSS IDs: 365.279, 365.280, 320003
Summer Semester

Here you get the **grade for the Bachelor's thesis**. Apart from handing in the thesis, there is no report or presentation needed.

Prequisit
- Seminar in AI (BSc) and Practical Work in AI (BSc)

Lectures
- It is mandatory to watch these lectures at least once

Work load
- 9 ECT: ~225h (~6 weeks full time)

Grading:
- Thesis Grading Guidelines
- No presentation or defense.
- There is no deadline for the thesis (I think the institute record was 6 years for an MSc thesis).
- Best practice: Create the story summary early on (see this section).

Other notes:
- If you do the Bachelor's Thesis at IML, you only have to be signed up for this KUSSS course. To get the grade, read the "Hand-in and grading section".
- Additional information is on this JKU page. Also see comments below on the general overview for master's students.

Seminar in AI (MSc)

KUSSS IDs: 365.204, 365.205, 365.377
Winter Semester

Prequisit

- This is a 3rd-semester course; we expect to have done most of the following (or similar) courses
    - ML: Supervised & Unsupervised Techniques
    - Math for AI I, II & III
    - DL: Basic, Architectures and Generative Techniques & Geometric Techniques
    - LSTM & Transformers
    - Computer Vision
    - Deep Reinforcement Learning

Goals

- Get started/practice reading papers
- Establish connections between publications
- Learn some basics for scientific presentations/reports (e.g., cite properly)

Lectures

- It is mandatory to watch these lectures at least once

Work load

- 3 ECT: ~75h (~ 2 weeks full time)

Output

- **Report**
    - **Contents:** Story summary, paper summary, and critical analysis.
    - **Length/structure:** Instructions and word limits in grading guidelines**.**
    - **Draft Deadline:** Two weeks before the presentation. Expect feedback within 2 weeks (send a polite reminder if needed).
    - **Final Deadline:** A month after the presentation.
    - **Final Submission:** see file uploads
- **Presentation**
    - **Format: In-person** (Linz or Vienna). Virtual only by approved email exception (e.g., students based in Bregenz).
    - **Task**: Present a paper plus your critical assessment.
    - **Resources:**
        - resources section, for example presentations, PowerPoint templates
        - grading guidelines, for instructions
    - **Time:** 15 mins + 5 mins Q&A. Practice to stay ≤15 min. TODO!!
    - **Registration: Send your report draft** (with story-summary questions answered) to your supervisor for permission to present at least 2 weeks before. Expect a reply within 5 days (send a polite reminder if needed). After **approval**, register for the presentation (link below) at least 1 week in advance.
    - **Preparation and feedback:** Send the slide draft to your supervisor at least 1 week in advance; expect feedback at least 2 days prior (send a polite reminder if needed).
    - **Peer review:** You must attend all presentations in your session and provide feedback to the presenters.
    - **Submission:** Upload slides by **23:59** the day before (see file uploads).

Grading

- Seminar Grading Guidelines.

- Once you have given your presentation and submitted the report, fill out the grading request. The grade should be available in KUSSS within a month.

Dates WS26
IMPORTANT: Please reserve the whole time range until your specific time slot is fixed.

- 1st Seminar presentation date: **Friday 4 December 2026: 09:00-19:00** in person in Linz/Vienna
- 2nd Seminar presentation date: **Thursday 11 March 2027: 09:00-19:00** in person in Linz/Vienna.

Practical Work in AI (MSc)

KUSSS IDs: 365.206, 365.207, 365.378
Officially only in the Winter Semester, but the start is flexible if your supervisor agrees.

Prequisit
- This is a 3rd-semester course; we expect to have done most of the following (or similar) courses
    - ML: Supervised & Unsupervised Techniques
    - Math for AI I, II & III
    - DL: Basic, Architectures and Generative Techniques & Geometric Techniques
    - LSTM & Transformers
    - Computer Vision
    - Deep Reinforcement Learning

Lectures
- It is mandatory to watch these lectures at least once

Work load
- 7.5 ECT: ~188h (~5 weeks full time)

Grading
- Practical Work Grading Guidelines
- No presentation!
- Report
    - The word limit is in the grading guidelines
    - Report deadline: End of the subsequent semester.
        - E.g., if you start in SS25, the report is due at the end of WS25 (i.e., around 1st of February 2026). Semester end dates are taken from here
        - Once the report and the grading request have been submitted, the grade should be available in KUSSS within a month.
- Code: either add a link to a GitHub repo or upload a separate zip file. Please don't put the code together with the report into a zip. We would really like to have the report as a separate PDF.

Master thesis

We (the SPT team at IML) do not organize master's theses, nor do we enter grades.

- The grade has to be entered by the supervisor. If the supervisor is confused, tell them there is another document for supervisors where they can look these things up. ☺
- Please add the story summary in Appendix A

We created this new "Road to Master" document. Please leave comments there for specific questions.

Work load

- 21 ECTS: ~525h (~13 weeks full time)

Master's Thesis Seminar (Poster Session)

KUSSS IDs: 365.253, 365.364, 365.365, 365.337
Both the Winter and Summer Semesters

For the Master's thesis seminar, a poster has to be presented at the poster session.

- Master's Thesis Seminar Grading Guidelines
- You may (but don't have to) use one of the templates from the templates section
- Explanation about Poster sessions at conferences
- Additional information is on this JKU page. Also see comments below on the general overview
- IMPORTANT: Please bring
    - Printed poster (A0, landscape; paper print is fine)
        - Students can print it at the Photocopying services: Services Department | JKU Linz (A0 and "Normalpapier" are fine; costs 5,50€ by 06/2025)
        - Time to print @ 'Kopierstelle': 1 working day.
    - Pins (DE: Nadeln) for the pinboard
- Please include your name, the name of the institute, and the name of your supervisor on your poster.
- Send the draft for the poster to your supervisor at least 1 week before the presentation. You should get feedback at least two days before the presentation. Feel free to send a friendly reminder in case it is necessary.
- **Poster upload:** at the latest at 23:59 the day BEFORE you present the poster (also see the uploads section). This is only necessary for students supervised by the IML (e.g., the CP institute might have its own rules for the poster upload).

You will have to present the result of your master's thesis in a poster session organized by the IML. It's okay if you have not fully completed (nor handed in) the master's thesis yet. But you should be able to clearly demonstrate your hypotheses, experiments, visualization of results, interpretation, and conclusion. Ultimately, ask your supervisor if what you have is presentable.

Lectures

- It is mandatory to watch these lectures at least once

Grading:

- Your poster presentation will be graded for this course (not the thesis)
    - You will be graded by us, not by your supervisor.
    - The master's thesis grade will be a separate grade (AFAIK 21 ECTS)

Upcoming Dates:

- **Time**: Tuesday, 2 February 2027 16:00 - 18:00
- **Place:** Kepler Hall (in person only!)
- **Registration:** fill out your details in this registration sheet at **least 10 days** before the poster session! Please also do this if you are supervised by another institute (e.g., CP institute), as we need an estimate for the number of participants.
- The poster session is open for anyone to join, so you can bring colleagues if they're interested.

Hand-in and grading

To receive a grade, you have to do a file upload (for reports/theses/posters) and a grade request. Only then can we grade you.

File uploads

In addition to the file uploads, please also send the report to your supervisor via email (or their preferred way of communication).

- NOTE: Generally, avoid sending raw PDFs if they are larger than 50 KB. Preferably send large documents with a cloud link (e.g., Google Drive).

Starting with SS26

If you are enrolled in the KUSSS course in SS26 or later, upload it via Moodle; otherwise, see the Nextcloud upload below.
Upload your document in the Moodle course. Choose the semester in which you enrolled in KUSSS for the upload. Before you can upload, you must do the quizzes for the lecture content.

For uploads in WS25 or older

We used our Nextcloud instance with this upload link.

File naming convention:

Please carefully stick to the file naming convention:

- <year-semester>-<Matriculation_number>-<Lastname>_<Firstname>-<report>-<version>-<short title>.pdf

The year/semester is NOT the registration semester, but the hand-in year/semester.
For example:

- 24SS-K1234567-Radler_Andreas-Thesis_MSc-v1-PIRADS_Score_Prediction.pdf

Here are the details

- For the **<year-semester>**: e.g., 24 for the year 2024. Use SS for the summer semester, and WS for the winter semester
- For **BSc,** the **<report>** has to be one of the following
    - “Seminar_Presentation_BSc” for the presentation
    - “Seminar_Report_BSc” for the report
    - “Practical_Work_Report_BSc”
    - “Practical_Work_Report_BSc_Code.zip” (for the code as a separate file)
    - “Thesis_BSc”
    - “Thesis_BSc_Code.zip” (for the code as a separate file)
- For **MSc,** the **<report>** has to be the following
    - “Seminar_Presentation_MSc” for the presentation
    - “Seminar_Report_MSc” for the report
    - “Practical_Work_Report_MSc”
    - “Practical_Work_Report_Code_MSc.zip” (for the code as a separate file)
    - “Poster_MSc”
    - “Thesis_MSc”
    - “Thesis_MSc_Code.zip” (for the code as a separate file)
- For the **<short title>,** try to summarize your title into max. 3 words. Reason: It’s just nice sometimes to search for something by name when we do the grading.
- The **<version>** is just counting up starting with v1, e.g. v1, v2, v3, etc.

Grade-request

Note: For **Master's Thesis**: Don’t submit a grade request, but please see the overview in the Master's Thesis section below (link). **For all other courses, please fill out a grade request**:

1. After you have done all tasks necessary to get your grade, please let us know you want to have a grade by filling out this form: grade request form.
2. **IMPORTANT:** After you have submitted this grade request form, contact your supervisor and ask them to enter the grade.
3. If there was any mistake in your request, let your supervisor know to delete the entry and resubmit a new grade request.



Why submit the grade request? Previously, we let supervisors fill out the “metadata” of a grade. This didn’t work well, as information was incomplete and mistakes were made. So we decided that students should fill it out and make sure it’s correct. Afterwards, supervisors give their grade. Finally, the grades are entered batch-wise into KUSSS by us.

Grading Guidelines

You can find the grading guidelines for all courses here.

FAQ

How to read a paper?

See this excellent advice:

- How to read a paper (short read)
- Ten simple rules for reading a scientific paper (longer read)

How to write a report/scientific paper?

This advice is a brief summary of the excellent book "**The Scientist's Guide to Writing**".

The story summary (important)

Answer the following questions and put the Qs and As after the abstract of your (SE/PW) report (for theses which are public documents, please put them into Appendix A)!

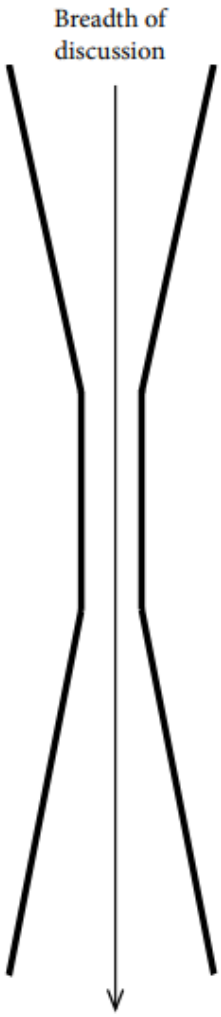
1. What is the central question?
2. Why is this question important?
3. What evidence/results are needed to answer this question?
4. What methods are used to get this evidence/results?
5. What analyses must be applied to the raw evidence/results to answer the central question?
6. What evidence/results were obtained?
7. What were the results of the analyses?
8. How did the analyses answer the central question?
9. What does this answer tell us about the broader field?
10. *Did the paper answer the question satisfactorily? Why (not)? (only relevant for the Seminar)*

Try to answer these questions in <= 3 sentences. If you know the answers to these questions, the writing will be much easier and faster for you. Also, more clarity will help your readers.
I recommend doing this early on, as well as before, while, and after writing the report.

This seminar report contains an excellent story summary.

The "hour-glass" metaphor

To reach a broader audience, try to follow the hour-glass structure (see image below). I.e., start and end broader. Go most in-depth in the methods, results, and discussion parts.



Front Matter
- *Title*
- *Byline and author affiliations*
- *Keywords, word counts, and other details*

Abstract

Introduction
- General context of the work
- Narrower research area and statement of its importance
- Identification of a gap or other need for research
- Specific research question meeting the identified need
- Summary of approach to answer the research question
- Announcement of principal findings

Methods
- Materials, species, field sites, mathematical techniques, etc. used in the research
- Observational/experimental procedures followed
- Methods for analysis of data

Results
- Results of observations, experiments, or modeling in text, tables, and/or figures
- Comparisons among results (e.g., observation vs. theory or treatment vs. control)

Discussion
- Interpretation of results to answer research questions
- Consideration of possible weaknesses
- Relationship of results to previous literature and broader implications of having answered research question
- Prospects for future progress

Back Matter
- *Acknowledgements*
- *References cited*
- *Appendices or online supplements*
- *Data archives*

Citation Style

Regardless of your choice of text editor (we recommend Overleaf with LaTeX), we advise using a citation style that is equivalent to the default citation style at leading ML conferences. This style corresponds to using the natbib citation style in LaTeX. It can be tested here by replacing line 8 with the following style options:

```
\setcitestyle{authoryear,round,citesep={;},aysep={,},yysep={;}}
```

Please use textual and parenthetical citations correctly. In LaTeX, they are obtained by `\citet` and `\citep`, respectively. The following examples illustrate the use cases for these two kinds of citations:

- **Textual**, i.e., the citation is integrated in the sentence: Einstein (1905) argues that energy and mass are related concepts.

- **Parenthetical**, i.e., the citation is not integrated in the sentence: Energy and mass are related concepts (Einstein, 1905).

How to give scientific presentations

- **Audience and their background: pick them up where they are and think first about what you can deliver to them**
  Speak their language; convince them of the relevance; adapt the level of technicality of the content and the visual aids (maths vs graphs vs images and hard facts vs storytelling);
- **Slides as your visual aids:**
  At most 1-2 points or steps of reasoning per slide; no dense, multi-line text blocks; only big fonts; avoid complete sentences - not needed since you speak them; the slide should reflect the logical structure of your presentation but not every piece of information; pictures and graphs can be very helpful: make them big and make sure you properly explain what is shown and why it is important to look at it
- **Time: convey as much as possible - but not more!**
  Roughly 1-2 minutes per slide; make sure you know how much time you have for the actual and how much time for the discussion after the talk
- **Presentation structure:** largely follows the structure of the seminar reports and theses; see this section for an overview
- **No reading, speak freely**
  Make sure that you formulate sentences at least mostly on the fly during the presentation; otherwise, you will talk much faster than you naturally do; cards and notes are ok if you need them, but you should mostly look into the audience.
- **Rehearse and make sure you understand everything you present**
  Practice will build up your confidence and will help you to speak fluently and to stay on time; this allows you to focus on the content and not so much on thinking about the right words

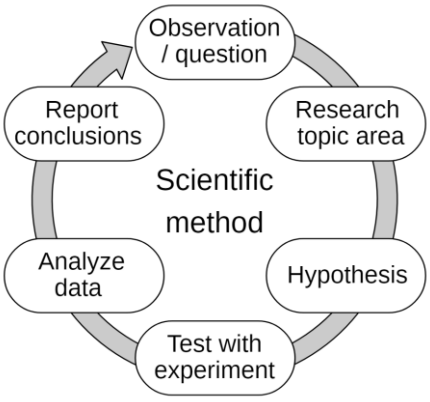
- **Be 100% aware of what you are saying:** Do not uncritically repeat what was said/written elsewhere; in case you are not entirely certain of the meaning of some tangent or related concept/work, please mention that - this is a sign of confidence
- **Prepare for discussion:** Identify expected questions from the audience; you may support your answers via back-up slides; not knowing the answer to a question is not a big deal; do not feel attacked or criticized by questions - audience engagement is almost always a sign of interest

Recommended further reading: link

Summary of the scientific method

The following steps usually follow a bigger "**research question**" which is in turn guided by an even bigger "**research agenda**". E.g., if you are interested in language modeling, the bigger research agenda might be "how to make LLMs work on mobile devices". Then a sub-research question might be "how can we do efficient knowledge distillation of LLMs?". Guided by such a research question, you might follow the process:

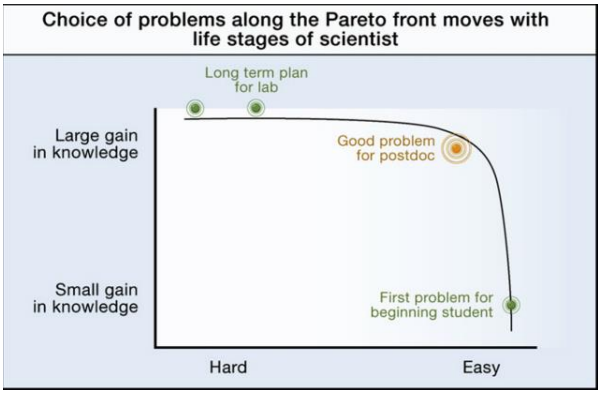
1. (1) First, you start with understanding the literature and the methods from a theoretical perspective.
2. (2) Then you formulate a hypothesis ("e.g., method A works better than method B on dataset X and Y").
3. (3) You do an experiment and collect data to validate your hypothesis (i.e., you implement the methods, run experiments, and collect different metrics, etc.).
4. (4) Then you evaluate and interpret the data.
5. (5) You might repeat steps 2-4 several times until you reach a conclusion. You write everything up – et voila!



Please watch the recording of Scientific Thinking, Working, Writing, and Reading. As mentioned there in more detail, usually steps 2, 3, and 4 are highly non-linear, and you often jump around

between them. Most (semi-)exploratory processes are highly non-linear. But it's VERY important that for EVERY experiment you do have a hypothesis in mind. Ask yourself, "What information do I want to get from this experiment?" Just randomly starting runs is a waste of resources and leads to confusion. Having a clear hypothesis in mind leads to more clarity.

Some tips regarding the thesis

- **Aim low and extend if you are successful:** The classic mistake is that people aim too high at the beginning. This should be avoided. Also see the following figure and read this article if you are interested (How to choose a good scientific problem, Uri Alon).
- 

- **Iterate fast**: Especially for experimenting, start small. Choose a MINIMUM VIABLE EXPERIMENT (similar to a Minimum Viable Product). For the first experiments, in the best case, you have a toy dataset with a small enough model and dataset that lets you train a model in a few seconds. When you fix all bugs, fully understand what's going on, tested your main hypotheses, you may generalize your codebase to a bigger dataset and test there.
- **Ask for help early:** Don't wait weeks to report to your supervisor. Aim for weekly or bi-weekly meetings (except for holidays).
- **Take breaks and holidays**: Take sufficient breaks and holidays to keep your mind fresh.
- **Don't let a break (holiday) lose your thread**: The scientific method is cognitively exhausting. A lot of pieces need to fall into place. The problem that often happens is that you take a longer break, forget some things (believe me, this happens to everyone easily), and then it's hard to get into it again. So be careful there. Try, in the best case, not to take a big holiday in between the (MSc) thesis.

- TODO: Do fellow students have more tips to add here?

Please discuss all these things with your supervisor. As supervisors tend to be busy, you might want to explicitly raise these points with them.

Thesis in industry, with another institute, or your own topic

General requirements:

- **Supervisor from JKU:** You need a supervisor from JKU with an AI-related background who wants and can supervise you. This is also the person who will decide on your grade. In general, it's not so easy to find a supervisor who did not offer the topic you want to work on in the first place. For a good approach, see the steps below.
- **Publishability**: A **master's thesis** needs to be publishable (make sure this is not against the company policy). The Master's thesis publication can be delayed for 5 years (Form: https://www.jku.at/fileadmin/gruppen/381/PAS/Abschlussarbeiten/Masterarbeit/Sperrantrag.pdf, Details:

  > If your master thesis or diploma thesis is to be excluded from publication, you must obtain the confirmation of the supervisor, fill out the "Sperrantrag" and send both by email to **hochschulschriften.bibliothek@jku.at** ↗. According to 86 para. 4 UG, the blocking request will only be granted if you can credibly demonstrate that important economic or legal interests are at risk.

  Source: https://www.jku.at/en/degree-programs/students/theses-and-dissertations/). A **bachelor's thesis** does not need to be publishable (to the general public), BUT it will be "einsehbar" (accessible) from the University. In both cases, you don't need to publish the data or the code, but the thesis. Clearly, you will also need to make plots, analyze your data, etc.
- We generally don't sign NDAs! However, the supervisor often does not need to know the confidential information to provide good supervision.

One good procedure (there are many) to approach this:

1. Clarify with your company whether they would want to do this. If it's a master's thesis, make sure it's publishable.
2. Find a supervisor from an institute: This is usually not straightforward.
    a. Search candidates among other institutes, and write them an email.
        i. If you find a supervisor at another institute AND they offer theses for AI studies, then you may stop here.
        ii. In case they don't offer theses, and the supervisor from the other institute wants to cooperate with us.
            1. If the supervisor has an AI background (very hard to say a priori; e.g., published >= 1 paper in typical AI conferences) or if you are not sure, please write me an email with a short description of the project and the name of the supervisor. Also, put the supervisor in CC. We will then decide on a case-by-case basis.
            2. If the supervisor does not clearly have an AI background, you need a second supervisor from our institute. In this case, proceed with "search candidates in IML".
    b. Search candidates in IML. Those are people who might share the same interests and have the skills and background to supervise you. For the interests of the supervisors, check the topic list, their paper list, or their institute's web page. Then write them an email. In the best case, you can stop here.
    c. The "looking for a supervisor email": If you think you have an interesting topic, you can also write a summary of that topic and write an email to radler@ml.jku.at. I will

then send it to the whole institute. If you are lucky, a potential supervisor will contact you.

   d. The "slippery slope" approach: Start with the seminar and practical work with a supervisor who does similar things to what you want to do. But accept his topic. After the seminar or the practical work, when you get to know each other better, you can maybe convince him/her better to switch to the topic you want. Or you may settle on something that is at the intersection of both of your interests.

3. Write your thesis
   a. The process is the same as for the all thesis (see "overview for master students" section). Only that your research question is specific to a company use case.
   b. Note: A master's thesis is still a first attempt at a research project, with a research question, hypotheses, experiments, analyses, etc. Make sure it's not "I'll just do my normal work and sell it as my master's thesis".
4. Thesis protection
   a. You may also "protect" the thesis for some years before it's published. This may be desirable for the company to keep "company secrets" still secret for longer
   b. See https://docs.google.com/document/d/1p5d0eykqaw0dfB-L62kPOpdetn8X3CgcJQ3wh7RhPlw/edit?tab=t.0#bookmark=id.1lqjxdwzp7sx in this document.

Is any presence required for the Seminar, Practical Work, or Thesis?

The seminar presentations will be in person.

For the practical work and thesis, it's ultimately the supervisor's choice. But most are extremely flexible, so virtually nothing should be a problem.

Use of Writing or Coding assistants (LLMs, etc.)

Any tool can be used to prepare high-quality reports/thesis.
BUT: The responsibility for the paper's entire content rests with the author; ensure that all content is accurate and original.

- Be aware of potential **plagiarism**: Ensure all sources and ideas are properly cited, and original content is clearly distinguished.
- **Carefully review** any LLM-generated content: A key grading criterion is that any LLM-generated text is thoroughly reviewed and edited.

Tools

- overleaf.com for LaTeX reports

    - You can work with the free account, where you can share the document with one person for free. This may be your supervisor to give you feedback directly on the document.
        - Note that on your free account, the compile time is limited.
    - As of Feb.2025, university staff now have an Overleaf Pro account.
        - So you could kindly (!!) ask your supervisor to create the Overleaf doc for you and share it with you. Then you can compile bigger documents.
        - Careful! Be nice – you are asking for a favor!
- As an IDE, I highly recommend PyCharm or VSCode
    - I recommend NOT using Jupyter for your PW or for the thesis. It's not only more cumbersome for the project itself. But also, there are many tools that IDEs provide which, if you are used to them, can facilitate your coding experience by A LOT. Also, in industry/research, everyone expects you to work with an IDE. The "single file" structure of Jupyter really brings in many limitations that make it not suitable for almost all projects.
    - Pro-Tip: Jupyter notebooks have their own advantages. For me personally, they are the right tool for fast development of small modules. Jupyter keeps the current execution context in memory. So you can put longer computations into a cell that you only execute once. Then you can focus on the logic you actually want to implement. Also, for creating visualizations of parts of a module, it's great.
- Compute
    - My advice for computing is: try to find a dataset/benchmark/setting where you don't need large computing. Ideally, you can just run it on your laptop and get the result in seconds to a few minutes.
    - For development, you can often try things out on MNIST-like datasets. When it works there, you may go to something bigger (if you have, e.g., a local GPU).
- Literature management tools
    - Zotero (or Mendeley)
- An engineering notebook
    - Usually, a physical sheet where you log stuff in the development process (see also the great blog post up there)

Youtube Channels

- Yannic Kilcher
- AI explained